

Summary of insulator defect detection based on deep learning

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ABSTRACT

Insulator is a kind of key electrical equipment in transmission line. Its long-term operation in strong electric field, high temperature sunshine, mechanical stress, humidity, dirt and other complex environment. This is likely to lead to self-detonation, pollution, flashover and leakage current and other defects, which will reduce insulation performance and seriously affect the safety of power grid. Therefore, power insulator defect detection is an important task in power system detection. How to identify insulator defects quickly and accurately has been an important problem for many scholars. Firstly, this paper reviews the insulator defect detection technology based on traditional detection algorithm. Secondly, the application and key technologies of deep learning algorithm in insulator defect detection are introduced in detail. Then it introduces the challenges and future development trend of this field; Finally, it summarizes this paper and puts forward some suggestions for the development of this field.

1. Introduction

With the continuous expansion of the scale of the power grid, the transmission line erection area continues to expand [1]. In order to make the transmission line work stably for a long time, it is necessary to carry out regular detection and check whether there are any defects in the components [2]. Patrol inspection of transmission lines to obtain image information of insulators, towers [3], hardware [4], conductors [5], grounding devices [6] and other components is to achieve status detection and fault diagnosis of transmission line equipment. As one of the most important components on the transmission line, the insulator has good mechanical support and electrical insulation performance. It plays an important role in supporting the conductor and preventing the current from returning to the ground during the entire transmission process [7]. Its normal operation directly affects the operation of the entire transmission line. Due to its long-term exposure to natural environment, it is easy to be disturbed by temperature, rain and snow, birds and other disturbances, resulting in self-explosion, damage, flashover and foreign matters [8]. The insulator is also the component with the highest probability of failure in the circuit, and failure of the insulator may affect the operation of the entire power system, leading to large-scale power outages, which will cause huge economic losses [9, 10]. Therefore, it is very necessary to detect and replace defective

insulators in time [11,12].

The earliest defective insulators are generally detected by manual or robot detection. It requires the inspectors or robots to go to the site to observe the appearance of the insulator and measure its temperature [13]. It not only consumes a lot of human resources, but also is easy to cause missed inspection and mis-inspection [14]. At the same time, there are security risks, slow detection and other problems. Early detection of insulators is generally based on traditional image processing and infrared imaging [15]. Although these methods solve the problems of low detection efficiency and high risk in manual detection, it has large limitations and is easy to be interfered by the environment, and can only be used in specific scenes. Related categories of insulator detection are summarized as shown in Fig. 1 below. It mainly includes detection methods, insulator types, insulator defects and insulator characteristics [16].

In recent years, the rapid development of deep learning [17] technology has promoted the development of insulator defect detection, and greatly improving the efficiency and accuracy of detection. At present, the technology combining UAV images and deep learning algorithm has become the mainstream insulator defect detection method, but there are still several factors restricting the development of this field.

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- Firstly, Complex background: The insulator is located in a variety of environments, including forests, farmland, buildings, rivers and mountains. In these complex backgrounds, it is undoubtedly a great challenge to accurately locate insulators and identify whether they have defects.
- Secondly, Different types of insulators: different insulators are made of different materials and have different characteristics and appearances, such as glass insulators, composite insulators and ceramic insulators, which adds difficulties to the detection process.
- Thirdly, The defective part of the insulator is very small: when the UAV is taking remote aerial photography, the defective part of the insulator is very small in the high resolution image, so it's easily missed detection.
- Finally, there are many types of insulators, and some defect characteristics are very small. The insulator is usually photographed by the inspection drone at a safe distance. The quality of the target image obtained is not high enough, which brings great challenges to the defect detection of insulators. All of these pose a great challenge to insulator defect detection.

2. Traditional insulator defect detection methods

Traditional detection methods required inspectors to use tools to closely observe the working condition of insulators, or to directly inspect videos or images of transmission lines taken by helicopters, a method that would not only consume a lot of time, but also a huge amount of data that would make the staff extremely prone to mis-inspection and missed inspections. For the inspection of transmission lines, there are mainly helicopter inspection [18,19], robot inspection [20,21] and UAV inspection [22,23], and for cost reasons, the current model is still mainly UAV inspection with other inspection methods as a supplement. Researchers obtain high-definition images of insulators through UAV inspection, and then determine whether insulators have potential safety problems by traditional target detection algorithms. There are many methods for insulator defect detection, including manual observation [24], Ultrasonic detection [25], Ultraviolet pulse method [26], Leakage current measurement [27] and Electric field method [28], As shown in Table 1.

- 1) Ultrasonic detection method: During the production process, the insulator may have defects due to accidents. With the long time operation in the transmission line, defects such as voids, delamination and interface separation may occur inside the insulator. Ultrasonic technology can be used to detect this kind of tiny defect inside the insulator. This method determines the location and type of the defect by receiving the acoustic signal inside the insulator and according to the signal difference of the defect area. Yuan et al. [29] completed the internal defect detection of insulators using ultrasonic phased array (UPA) technology, which can detect the types and locations of various internal defects. And the detection thickness is large, fast, low cost and harmless to human body. However, the method can only detect mechanical damage faults, and cannot detect degradation and abnormal corona discharge of insulators.

- 2) Ultraviolet pulse method: The aging of the insulator due to long working hours will lead to carbonization channels, erosion and cracks on the surface of the mandrel, which will lead to stable corona discharge of the insulator, further damaging the insulation effect of the insulator and causing insulation breakdown when it reaches a certain level, thus causing serious safety accidents. The UV pulse method is to detect the defective area based on the uneven distribution voltage and partial discharge on the insulator surface caused by the corona discharge at the abnormal location of the insulator. The detection distance of this method is long, and it has a good detection effect for the local corona caused by carbonization channel and etching damage. However, this method can only detect insulator corona discharge, and the rest of the fault detection is not obvious, while there is leakage detection and limited anti-interference ability. Liu et al. [30] detected defective insulators by using a UV detector to measure the abnormal corona discharge of insulators.
- 3) leakage current measurement method: Leakage current method is generally used to detect the insulator in which flashover occurs. When there is dirt on the insulator surface, its normal operation flowing through the wet dirt layer on the surface will generate leakage current, and when the dirt accumulates to a certain level, it will cause flashover on the insulator surface. So the insulator flashover can be detected by measuring the size of the leakage current on the insulator surface. Meanwhile, as the insulator deteriorates, the flashover voltage decreases and the leakage current increases, so measuring the size of the leakage current can also detect the degree of insulator deterioration. However, the leakage current method is subject to external influence and does not accurately reflect the degree of insulator degradation, and it requires installation of detection equipment on each insulator string, which is too costly. Du et al. [31] used recurrence diagram technique to analyze the leakage current on the insulator surface and extracted three different frequency components of the insulator flashing process, and concluded that the recurrence diagram of high frequency components can accurately reflect the insulator flashing situation.
- 4) Electric field method: At present, the electric field method is a conventional method for detecting defects occurring internally due to insulator degradation in transmission lines. this method measures the electric field strength on the insulator surface, initially determines the location of the defect based on the electric field simulation, and then accurately identifies the location of the degraded insulator defect by means of an electric field detector. Wang et al. [32] first analyzed the electric field around the insulator, and then used an electric field detector to accurately identify the degraded insulator. This method can accurately and intuitively determine the performance changes of insulators and is more resistant to interference, but the detection is more dangerous and is not effective for dirty and wet insulators. For cost considerations, the current detection of transmission lines are based on image detection, and the most common early detection methods are mainly divided into two categories, one is based on image processing [33], this method sets a special condition to distinguish insulators and defects from the background. In the past few years, the detection algorithm based on

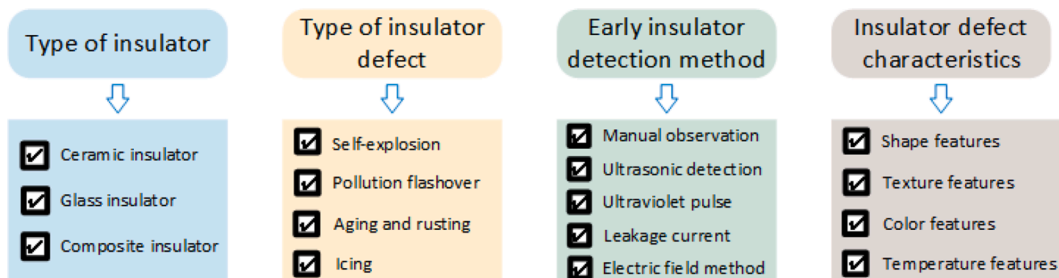


Fig. 1. A summary of insulator detection.

Table 1
The detection methods of insulators.

Methods	Advantages	Disadvantages	Application Scenarios	Principle of the method
Observation method [24]	Low cost and easy to use.	Low efficiency and high error rate.	Detection of insulator surface defects.	The inspection staff directly observes the insulator's appearance or measures its internal temperature by means of climbing or drones.
Ultrasonic detection method [25]	Detection of large thickness, fast, low cost, harmless to human body.	Only detects mechanical damage faults, single type of detection.	Detection of minor defects inside insulators.	This method receives the acoustic signal inside the insulator and determines the location as well as the type of defect based on the signal variability in the defect area.
Ultraviolet pulse method [26]	High sensitivity, low noise, wide linear range.	Sensitive to temperature changes, expensive equipment.	Detection of insulator discharge under high humidity conditions.	The UV pulse method is to detect the defective area based on the uneven distribution voltage and partial discharge on the insulator surface caused by the corona discharge at the abnormal location of the insulator.
Leakage current measurement method [27]	Can detect insulator flashover and degradation degree.	Cumbersome maintenance and expensive equipment.	Detection of dirt-accumulation on insulator surfaces	This method determines whether an insulator has a flashover fault by detecting the leakage current on the insulator surface.
Electric field method [28]	Accurate detection and strong anti-interference ability.	High risk, single type of detection.	Detection of deteriorated insulators.	This method determines the location of insulator defects by measuring the electric field strength on the insulator surface.
Traditional image processing method [33]	Simple operation and low cost.	Weak anti-interference capability and low detection accuracy.	Insulator surface characteristics defects.	The method works by monitoring the discharge on the insulator surface.
Infrared imaging method [37]	Simple operation and clear results.	Costly and susceptible to temperature.	Detects abnormal change -s in the internal temperature of the insulator.	This method identifies defective insulators by using instruments to detect local temperature changes within the insulator.

traditional images has been applied by more and more researchers to detect defective insulators by extracting the external contour features, shape, texture features and color features of UAV patrol images. For example, these typical algorithms such as contour detection [34], color feature detection [35], shape texture feature [36]. However, these algorithms require human-designed classification features, and due to the complex environment in which the insulator is located and the characteristics of the defective parts are not obvious, the accuracy and robustness of such detection methods are poor, and they are greatly disturbed by environmental factors, which cannot meet the accurate inspection of the defective insulator in the transmission line; The other is based on infrared imaging [37], which

uses infrared imaging method to detect the change of insulator internal temperature to identify defective insulators. Due to the influence of sunlight, the temperature difference between the Sun-ward side and dorsal side insulator may be caused, resulting in false detection. Therefore, the detection accuracy of this method is easily influenced by the light intensity, with certain limitations, and can only be used in cloudy days.

2.1. Insulator defect detection based on traditional image processing

At present, there are two kinds of methods to extract insulator

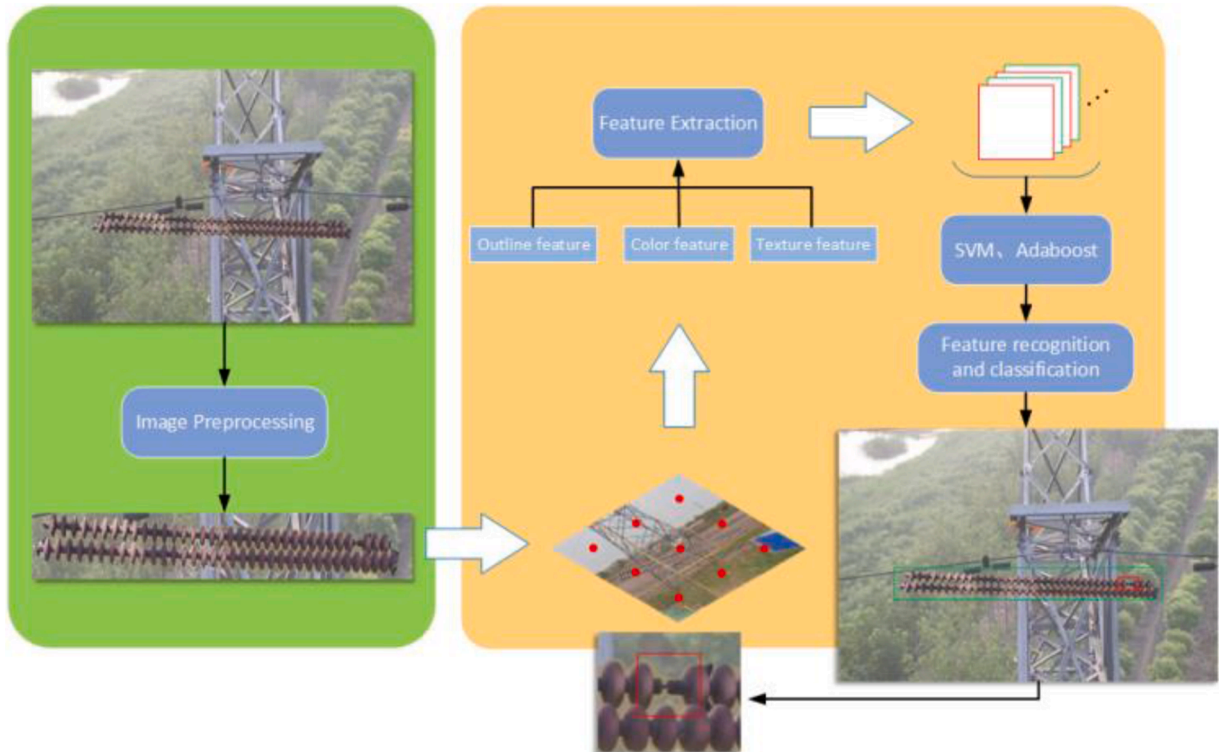


Fig. 2. Insulator defect detection based on image features.

features from aerial images, which are based on surface features and based on gray features. The method based on surface features is according to the remarkable difference between the shape of insulator and its surrounding environment, so its unique surface features can be used to achieve positioning and recognition. Yu et al. [38] proposed an active profile model to segment insulators according to the shape and texture characteristics of insulators, which proved that defective insulators can also be effectively identified in the background of clutter. Wu et al. [39] proposed a method to extract insulator texture features by using semi local operators under the Beltrami framework, which improved the processing ability of insulator surface nonuniform texture features. Fang et al. [40] introduced two factors of color and distance in the design of the algorithm, and optimized the algorithm using prior information, so that the algorithm can effectively avoid missing and false detection. The algorithm has fine detection effects on insulator loss, pollution flashover and bird droppings accumulation.

The working principle diagram of based the image feature method is shown in Fig. 2. First, image pre-processing is performed on the training samples, then Extract image features and classify them, and then results. Image pre-processing is to segment the target area in the image from the surrounding environment. Image transmitted to the classifier to output the classification segmentation is an essential step in the image pre-processing process, and is also one of the most difficult tasks. Zhang et al. [41]. proposed a k-means clustering method to segment insulator images in order to accurately locate insulators in complex environments. Feature extraction is to extract the profile, texture, color and other features of insulators in the image. It accuracy of classification. Ye et al. [42]. used the feature extraction method to separate the insulator from the is a key step in the image processing process, which directly affects the efficiency of classifier recognition and the background. Classifier recognition refers to transferring insulators to be tested into a trained classifier to detect, and then outputting the results of insulator defect detection. The accuracy of classification is the common goal of all classification algorithms. At present, the classifiers used for insulator defect detection mainly include support vector machine (SVM) [43] and AdaBoost [44]. Reference [45] uses SVM combined with wavelet multi-resolution analysis (MRA) to accomplish defect detection of transmission line insulators, capturing insulator images using MRA and then detecting the state of insulators by SVM. Wu et al. [46] combined an Adaboost cascade classifier with the region of interest to complete the detection of insulators.

The defect insulator detection method based on gray level feature is that because the gray level and color in the insulator image are different from the surrounding background environment, the insulator can be separated from the background using image segmentation method, so as to achieve accurate positioning and detection of insulator profile. Tan et al. [47] proposed a fusion algorithm based on ceramic chip profile features and gray level similarity matching, which not only realized the profile extraction and accurate separation of insulators, but also built an insulator defect detection model based on ceramic chip spacing and gray level similarity matching. Wu et al. [48] uses the gray level co-occurrence matrix (GLCM) to divide the extracted insulator features into two categories, one with strong discrimination and the other with weak discrimination. This method can segment insulators from simple and complex environments. Zhang et al. [49] proposed a method of insulator feature extraction based on computer vision. This method extracts texture features through gray level co-occurrence matrix, and then uses local features to detect insulator features.

Although the current insulator defect detection method based on image features has made great achievements, the effect of detecting defective insulators by image features is usually significant when the insulator surface changes significantly, and the effect of detecting the defect of local temperature rise caused by abnormal discharge inside the insulator is very small, and it is vulnerable to the interference of complex background, resulting in low detection accuracy.

2.2. Insulator defect detection based on infrared imaging

The method based on traditional image processing is easily affected by external factors such as weather and environment, which makes the captured image unclear, leading to low detection accuracy and failure to detect abnormal temperature changes inside insulators. Therefore, researchers began to use infrared imaging method [50] to detect defective insulators. For the detection of internal defects of insulators, infrared detection is the most commonly used and one of the most effective detection methods. The excessive concentration of the internal electric field caused by the defect of the insulator and the partial discharge of the insulator are the main reasons for the abnormal temperature of the insulator. At the same time, the surface of the deteriorated insulator will also produce excessive leakage current, and the leakage current is concentrated in the deteriorated part, resulting in low zero value, internal defects and other problems of the insulator [51], which is also one of the reasons for the abnormal temperature of the insulator. The use of infrared thermal image can detect the temperature distribution on the surface of the charged insulator at close or long distance with low workload and high operational safety, which can effectively determine the fault situation inside the insulator. However, infrared instruments are expensive and the detection results are easily affected by factors such as sunlight, humidity and ambient temperature as well as the ability to cause sharp changes in insulator surface temperature. Therefore, it is better to use infrared imaging method to detect defects in insulators when the temperature difference is not large and the humidity is low.

In order to solve the influence of complex background of substation on infrared insulator image detection, Zheng et al. [52] proposed an infrared insulator image detection model based on improved feature fusion single point multi box detector, which designed a new feature pyramid and feature enhancement module to improve the feature extraction ability of insulator infrared image. The above insulator defect detection method based on infrared image has greatly improved the detection efficiency compared with the traditional detection method. At the same time, it is less interfered by the complex background, but it is vulnerable to the impact of environmental temperature, resulting in false detection.

3. Insulator defect detection methods based on deep learning

At present, the detection of defective insulators in transmission lines is mainly realized by aerial photography of UAV. Compared with manual detection, UAV detection has the advantages of low cost, high efficiency and high security. However, because the UAV is always in motion, the captured insulator images may be fuzzy and unclear, which affects the training process of the network model and is not conducive to the subsequent insulator positioning and defect identification. In order to further improve the quality of transmission line inspection, many researchers have started to work on the identification and defect detection of power equipment in UAV aerial images [53,54]. For example, Zhou et al. [55] used a wide and deep super resolution network (WDSR) to improve the image resolution of insulators by four times, and then used the detection algorithm based on the feature pyramid network to identify and detect defective insulators.

Deep learning-based methods have shown great advantages in complex tasks such as image classification [56] and defect detection [57] since 2012, and their network structure generally consists of an input layer, a convolutional layer, a sampling layer, and a fully connected layer [58]. The biggest advantage of deep learning-based detection methods over traditional detection methods is that they are more accurate and take less time to detect. At the same time, it can extract deeper features directly from the original image autonomously, with stronger generalization and expression ability, and most importantly, the deep learning-based target detection method can reduce the influence of complex environment on the detection results, and is more flexible and adaptable to diverse tasks than traditional algorithms.

Nowadays, it is no longer practical to manually confirm one by one whether there are defects in insulator images taken by UAVs aerially, so many researchers have started to apply deep learning methods to insulator defect detection. In recent years, many target detection algorithms using deep learning based algorithms have achieved state-of-the-art results in the detection of transmission line insulators and largely outperformed traditional algorithms [59]. Fig. 3 shows the flowchart of insulator defect detection by UAV combined with deep learning algorithm.

At present, the mainstream target detection algorithms based on deep learning can be divided into two categories: based on candidate region (two-stage) and regression(one stage). The target detection algorithm based on candidate regions has the characteristics of complex structure, high detection accuracy but slow speed. This kind of algorithm first processes the input image to generate candidate regions through convolution, pooling and other operations, then classifies the information in the regions and modifies the coordinates, and finally outputs the results. Such algorithms mainly include: R-CNN [60], Faster R-CNN [61], FPN [62], Mask R-CNN [63], etc.; The regression-based algorithm has a simple structure, fast detection speed but low accuracy compared with the above algorithms. This kind of algorithm directly regresses the input image through convolutional neural network (CNN), and then outputs the final result. It mainly includes YOLO(you only look once) series and SSD(Single Shot multibox Detector)series algorithms. Their working principles are shown in Figs. 4 and 5 respectively.

3.1. Application of two stage algorithm in insulator defect detection

With the development of deep learning technology, target detection algorithms based on deep learning are also widely used in the detection

of defective insulators in transmission lines. Two stage algorithm first extracts candidate frames from the image, then detects the targets in the candidate frames and judges whether there are target instances of a given category.

3.1.1. R-CNN

The R-CNN algorithm was proposed by Girshick et al. in 2014. As the first model in two-stage target detection, the model first selects several proposed regions from the image, then marks their categories and bounding boxes, uses CNN for forward calculation, extracts features from each proposed region, and finally uses the features of each region to predict their categories and bounding boxes. Zheng et al. [64] used CNN based on the R-CNN to extract visual features from patrol images, and then completes the detection of insulator self explosion defects. This method can accurately identify the location of insulators in different environments and complete defect detection. However, R-CNN requires multi-stage training process, the training process is more complicated, and the image is prone to distortion, which is slower compared to other target detection algorithms. Its workflow is shown in Fig. 6. Specifically, R-CNN mainly consists of three parts:

- Firstly, Selectively search the input image to generate multiple suggestion areas.
- Secondly, CNN network is used to extract feature information of candidate regions.
- Finally, Take the features and labels of each area as samples to predict the target category and boundary box.

3.1.2. Faster R-CNN

Shortly after Fast R-CNN was proposed, Ren et al. Proposed Faster R-CNN to solve the problem of slow candidate box region extraction. In

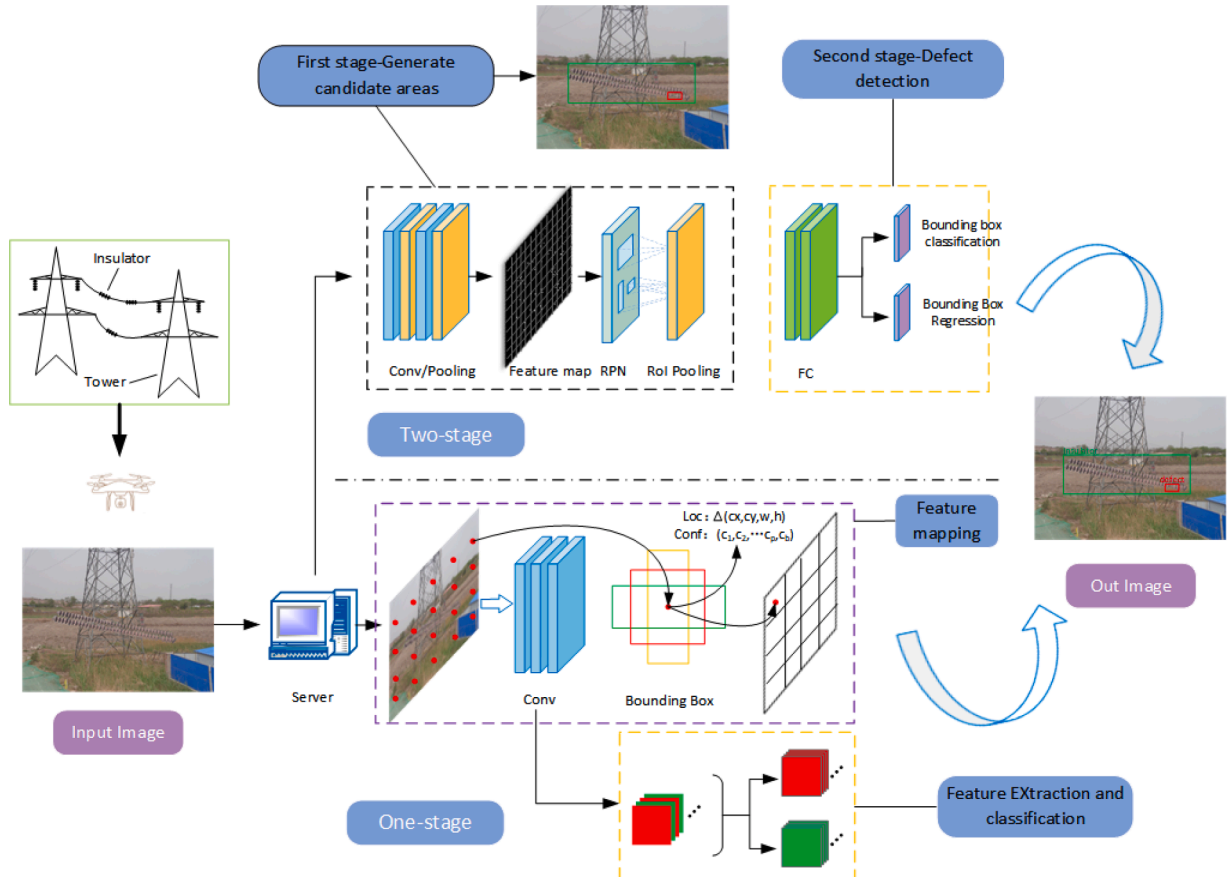


Fig. 3. Flow chart of insulator defect detection in UAV aerial photography based on depth learning algorithm.

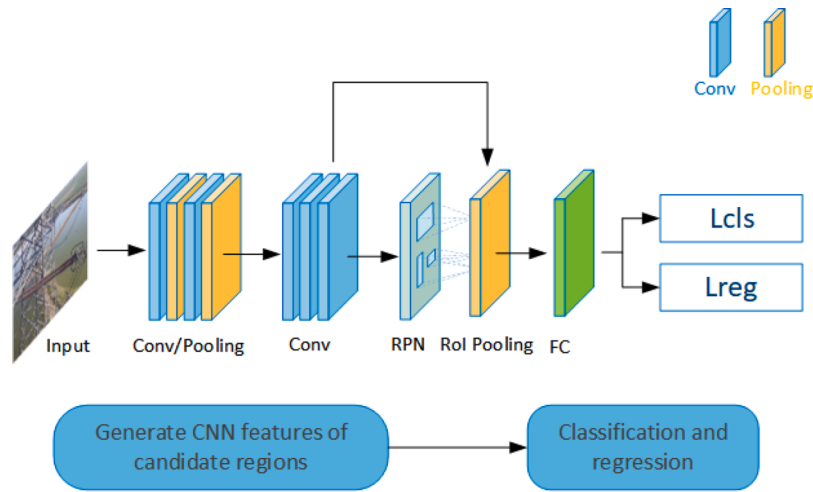


Fig. 4. Target detection algorithm based on candidate region.

this network, a region proposal network(RPN) is proposed for the first time to replace the original selective search algorithm. In this model, RPN has designed a multi reference window mechanism, which enables the model to complete candidate region recommendation, feature extraction, location and classification in the same network and generating candidate frames through convolutional feature maps, compared with Fast R-CNN algorithm, greatly improves the training efficiency and performance. In addition, Faster R-CNN is a target detection algorithm with end-to-end training and real time performance on GPU, As a mainstream target detection algorithm, it is often used to locate and identify defective insulators in transmission lines, but its detection effect for small targets is not ideal. Its network structure is shown in Fig. 7. On the basis of Faster R-CNN, Zhao et al. [65] has improved the anchor generation method and non maximum suppression (NMS) of regional recommendation network (RPN), which can effectively distinguish and detect insulator defects with occlusion. However, Wen et al. [66] proposed two Faster R-CNN based methods, Exact R-CNN and CME-CNN, to solve the problem of low detection accuracy caused by complex backgrounds and small targets, which can effectively detect the defects of insulators. To overcome the influence of complex background, Lei et al. [67] used Faster R-CNN algorithm to transform the insulator classification problem into insulator detection and identification problem, and the model can effectively identify defective insulators and bird nests, however, the speed of detection is slow and cannot meet the requirement of real-time detection.

3.1.3. FPN-Net(feature pyramid networks)

Lin et al. put forward the FPN algorithm on the basis of Faster R-CNN algorithm in 2017. This algorithm adds multi-layer features and feature fusion on the basis of the original network, which improves the shortcomings of Faster R-CNN algorithm in extracting semantic information

and greatly improves the detection effect for small targets. Hao et al. [68] used the cross stage and residual split attention (CSPNeSt) network as the backbone network, and added multi-scale bi-directional feature pyramid network (Bi-SimAM-FPN) to solve the problem of small target insulator defect detection.

3.1.4. Mask R-CNN

He et al. put forward Mask R-CNN in 2017, adding a branch that combines image semantic segmentation and target detection on the original basis. This model uses RoI Align layer to replace RoI pooling layer in Faster R-CNN. In order to avoid the deviation problem caused by integer quantization of feature map and original image in RoI pool, linear interpolation algorithm is added, This algorithm can make the features obtained by each receptive field better align with the receptive field area of the original image, thus improving the detection accuracy. In order to automatically identify insulators and their defects in the complex highspeed railway catenary, Reference [69] proposed an intelligent defect detection algorithm for catenary based on Mask R-CNN and image processing model, which can accurately identify insulators and their defects.

3.2. Application of one stage algorithm in insulator defect detection

The biggest difference between the one stage target detection algorithm and the two-stage target detection algorithm is that the former has no candidate region generation stage. This type of algorithm directly uses one stage to determine the target category and get the location detection box. The training process is also relatively simple. At present, the algorithms used in the field of insulator defect detection mainly include YOLOv3-v5, YOLOX algorithm and SSD algorithm.

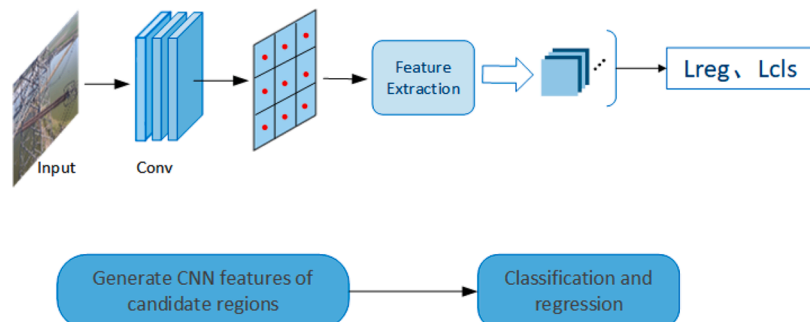


Fig. 5. Target detection algorithm based on regression.

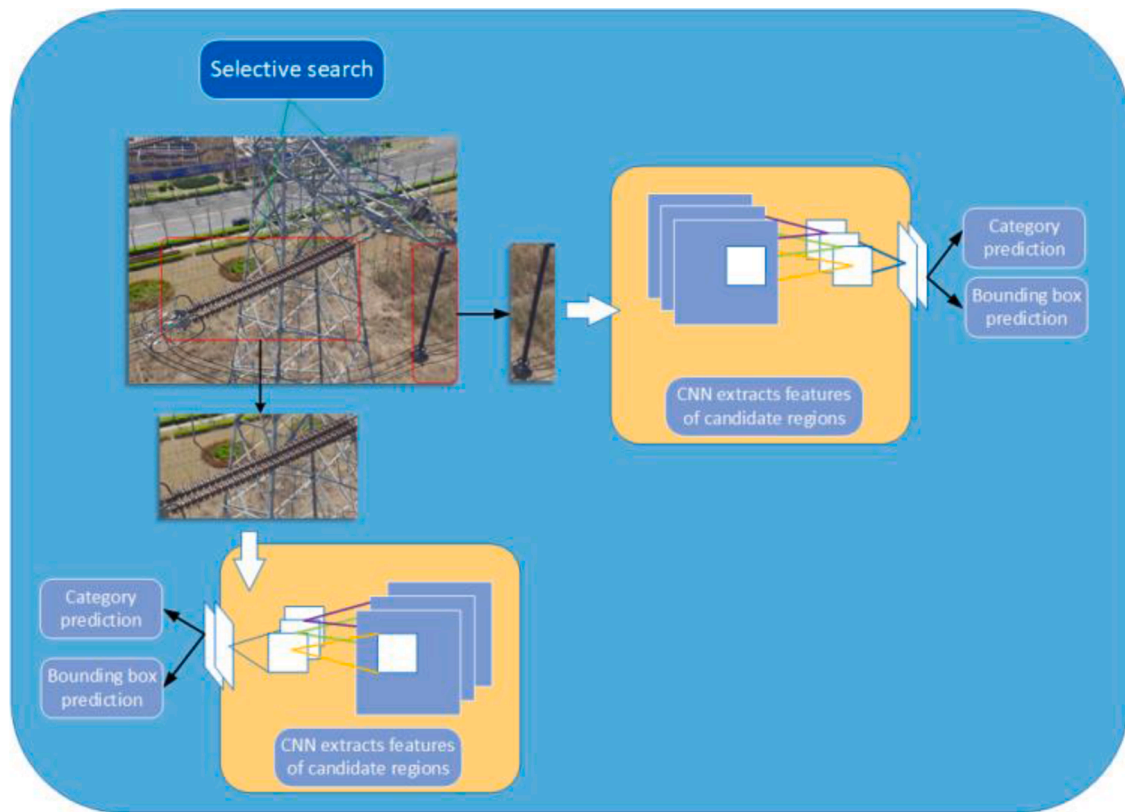


Fig. 6. Target detection algorithm based on candidate region.

3.2.1. YOLO series algorithm

YOLOv3 [70] model changes the darknet-19 of YOLOv2 model to darknet-53, and provides the tiny dark-net version. If you want to pursue faster speed, use tiny dark-net. The network also introduces the FPN structure to achieve multiscale prediction, and can achieve faster detection speed by changing the network model. Although YOLO series

algorithm has been updated to YOLOv7, YOLOv3, as a classic algorithm of YOLO series, has well balanced the relationship between detection accuracy and speed, so it is still one of the mainstream algorithms for insulator defect detection. The network structure of the algorithm is shown in Fig. 8. Reference [71] proposed a new CNN target detection method based on YOLOv3 network. By applying the latest EIoU and loss

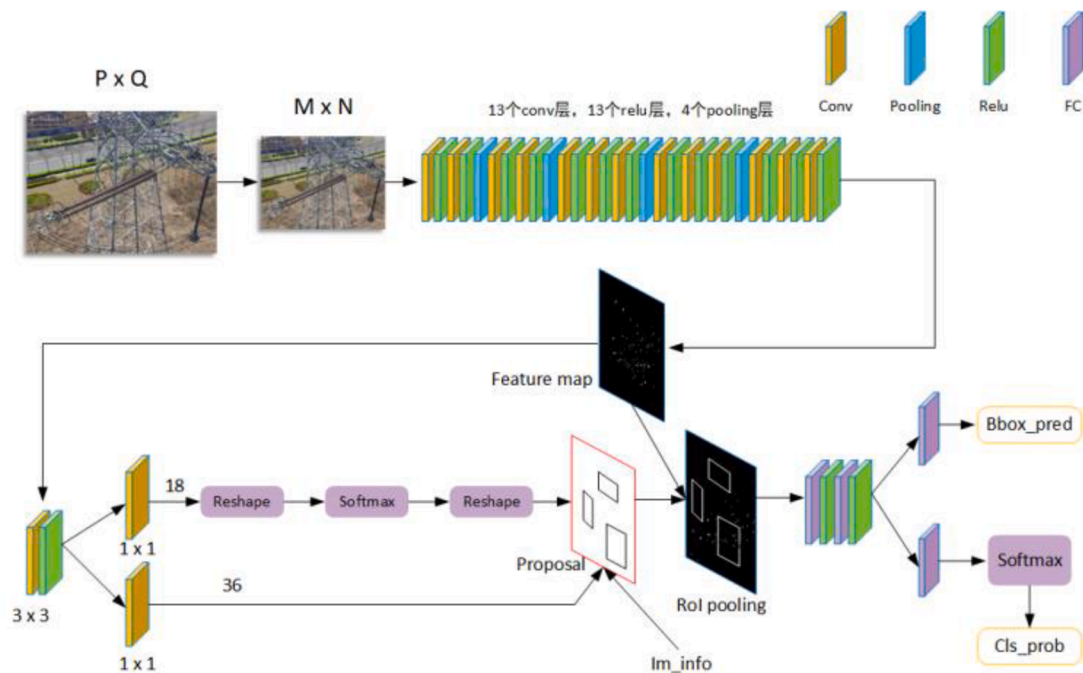


Fig. 7. Faster R-CNN network structure.

function to YOLOv3, the detection speed can be guaranteed while the detection efficiency can be improved. In order to solve the problem of small defect positions of insulators, difficult feature extraction, and overcome the interference of complex environments, Yang [72] et al. proposed an improved YOLOv3 algorithm to locate insulator positions in combination with spatial pyramid pooling (SPP) and Mobile-Net network, and then combined with Dempster Shafer (D-S) evidence theory to complete high-precision defect identification of insulators.

YOLOv4 [73] is a major breakthrough of YOLO series algorithms. This algorithm introduces CSPDarknet-53 as the backbone feature extraction network. In order to improve the image extraction effect, it also introduces SPP network, and selects Mish as the activation function. At the same time, it uses Mosaic method to enhance the data set, and uses label smoothing method to prevent over fitting and other problems. These improvements make YOLOv4 an extremely efficient and powerful target detector. Although YOLO algorithm based on deep learning has higher accuracy and faster detection speed in insulator recognition and defect detection. However, the performance of these networks is usually affected by shooting conditions and aerial images with different types of insulators and complex backgrounds, resulting in poor detection results. In order to solve these problems, Qiu et al. [74] first performed Grap-cut datasets enhancement on the datasets, then subjected the YOLOv4 model to Mobile-Net light-weighting, and finally improved the detection accuracy of the lightweight YOLOv4 model by Laplace sharpening. Reference [75] continued to make lightweight improvements based on Tiny-YOLOv4, a lightweight network of YOLOv4, to alleviate the problems of weak embedded computing power and low accuracy of model migration detection.

Shortly after the appearance of YOLOv4, YOLOv5 also appeared. Unlike previous generations, the algorithm was published by another research team, which did not have much innovation in the algorithm, and the team did not publish a paper. However, YOLOv5 significantly reduced the size of the model and greatly improved the operation speed, which led to some objections to its existence. According to the network depth and the width of the feature map, YOLOv5 can be divided into four versions: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. YOLOv5s is the network with the lowest depth and the smallest width of the feature map. As shown in Table 2, the performance comparison of the

four versions of YOLOv5 on the COCO datasets. Han [76] et al. improved the YOLOv5 algorithm by using the ECANet module to enhance the separation of background and target, while strengthening the algorithm for the detection of small targets. Bao et al. [77] adds coordinate attention (CA) to the YOLOv5 model and uses the bidirectional feature pyramid network (BiFPN) to replace the original PANet, so as to complete the detection of insulator defect site. Meanwhile, this method improves the detection accuracy for small targets. Zhou et al. [78] improved the YOLOv5 model in four aspects such as feature extraction, privileged fusion, candidate frame generation and loss function, and was able to achieve 98% accuracy for insulator defect detection.

YOLOx [79] algorithm was proposed and published in 2021 by Megvii. Similar to YOLOv5, YOLOx algorithm can also be divided into standard network structure and lightweight network structure according to different network width and height. From the perspective of anchor frame based algorithms, the author considers that YOLOv4 and YOLOv5 series algorithms may have some over optimization, and due to limited computing resources, software support is insufficient in practical applications, and YOLOv3 is still one of the most widely used detectors, so he selects YOLOv3-SPP algorithm as the benchmark model. Compared with the original YOLOv3 algorithm, this algorithm adds SPP components, which has better performance. Dai et al. [80] built a new method, YOLOD, by adding Gaussian functions to YOLOx detection head. This method solves the uncertainty of boundary box prediction during insulator detection and improves the robustness of detection. Han et al. [81] proposed an improved SiOU loss function based on the YOLOx algorithm, which accelerated the convergence of the model and introduced the ECA attention mechanism, which overcame the interference

Table 2

Comparison of YOLOv5 parameters in COCO datasets.

Network model	Image size	mAP Val 0.5	mAPVal 0.5:0.95	Detection speed/100/ms	Parameter (M)
YOLOv5s	640	55.4	36.7	2.0	7.3
YOLOv5m	640	63.1	44.5	2.7	21.4
YOLOv5l	640	66.9	48.2	3.8	47.0
YOLOv5x	640	68.8	50.4	6.1	87.7

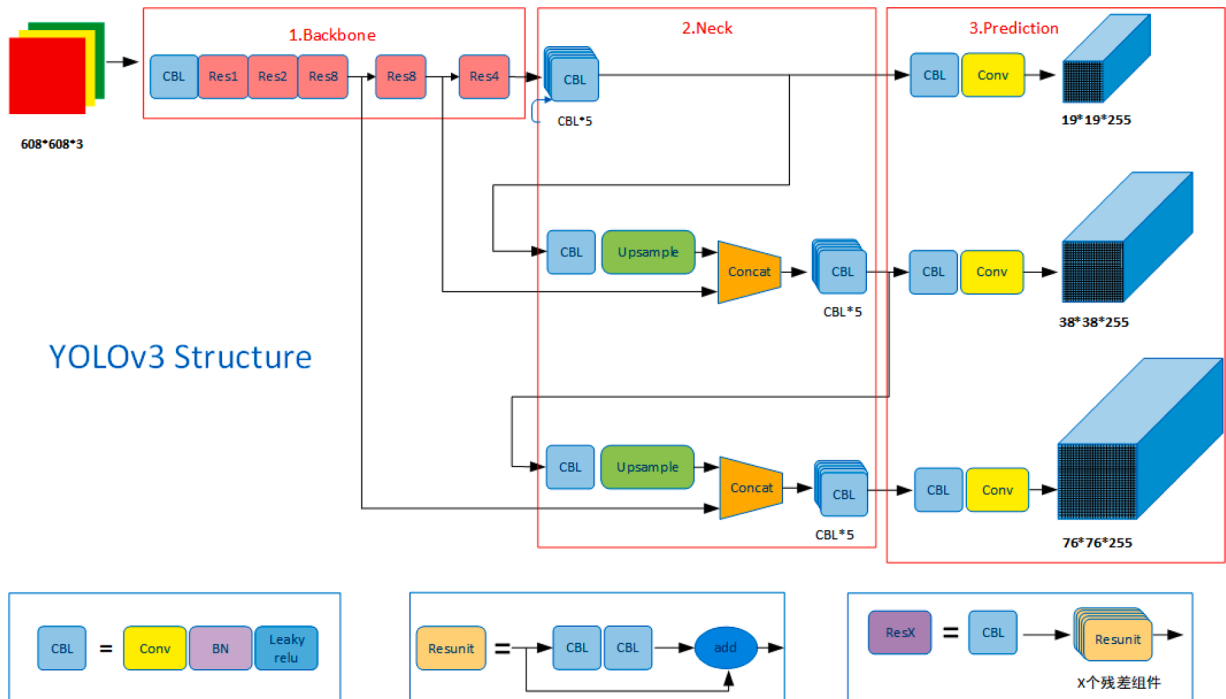


Fig. 8. YOLOv3 work flow chart.

of complex environment to a certain extent, and can accurately and real-time detect insulators and their defects.

3.2.2. SSD series algorithm

In 2015, Liu et al. [82] proposed SSD algorithm based on YOLO algorithm and Faster R-CNN algorithm, which combines the advantages of fast detection speed of YOLO and accurate positioning of Faster R-CNN. The main improvement lies in the introduction of multi reference and multi resolution detection technology, and the detection effect for small targets has also been greatly improved. Its workflow is shown in Fig. 9. Reference [83] used the method of combining SSD model and two-stage fine-tuning strategy to complete the detection of defective insulators. This method can automatically extract multi-level features of the image, and can quickly and accurately identify porcelain insulators and composite insulators under complex backgrounds. Wang et al. [84] first improved the SSD algorithm by adjusting the network parameters and the model optimization method, and then expanded the training sample through multi-scale training and horizontal mirror image method to improve the accuracy of model detection.

3.3. Insulator defect detection cases based on deep learning method

In order to verify the effectiveness of the deep learning based algorithm for insulator defect detection, the team conducted some experiments. By using the most classic algorithm of You Only Look Once algorithm, namely YOLOv5, the most common self-detonation defect detection of insulators was completed, and the detection speed reached 41FPS, achieving the effect of real-time detection and high detection accuracy. Compared with the general algorithm, the detection accuracy has some advantages. The following Fig. 10 shows the monitoring results. It can be seen that insulators can be identified with high confidence.

The comparison between the traditional method of insulator defect detection and the depth based learning method is shown in Table 3. Among them, references [45–59] are conventional detection methods, [74–79] are two-stage detection methods, and [81–90] are one-stage detection methods.

4. Insulator data sets and defect types

4.1. Types of insulators

Because the materials used by insulators are different, they have different characteristics. Although the structure and appearance of different types of insulators are quite different, they are both composed of insulators and connecting fittings. Choosing an appropriate method for defect detection is conducive to improving the accuracy and speed of detection. The following are some common insulator types:

Ceramic insulator [85] : insulator made of electrical ceramics. Electrical ceramics are baked from quartz, feldspar and clay. The ceramic surface of ceramic insulator is usually covered with ceramic glaze to improve its mechanical strength, prevent water infiltration and increase the surface smoothness. Among all kinds of insulators, ceramic insulators are most commonly used. After a long period of operation, both its insulation and mechanical properties will deteriorate and degrade to a certain extent, and eventually lose their insulation properties completely. In order to detect deteriorated porcelain insulators in a timely manner, Liu et al. [86] first extracted the infrared images of insulators, and then optimized and trained the model using a convolutional neural network LeNet, which is able to detect deteriorated insulators with high accuracy and robustness even under different conditions. Yang et al. [87] takes the UV image of ceramic insulator as the research object, establishes the UV image sample library under different states, uses AlexNet to train at different learning rates, and this method realizes the autonomous feature extraction and high-precision defect assessment of the UV image of porcelain insulator. Reference [88] first reconstructed the training images to expand the data set using the super-resolution network of the deep Laplace pyramid, and then corrected the training images significantly using low-light image enhancement techniques to improve the detection accuracy of porcelain insulator defects.

Glass insulator [89] : insulator made of tempered glass. Its surface is in the state of compression pre-stress. If cracks and electric breakdown occur, the glass insulator will break into small pieces by itself, commonly known as "self explosion". For glass insulator, it has zero value self explosion property, mainly because there are some impurities and nodules in the production process of glass insulator, and it is very easy to self explode under the influence of external load and temperature difference between cold and hot during operation, which also makes it unnecessary to conduct "zero value" detection for glass insulator during operation. Cheng et al. [90] segmented the image according to the special color features of glass insulator, and then used the spatial features contained in the aerial image to detect and locate the self exploding glass insulator. Gao et al. [91] used YOLOv4 algorithm to extract the deep features of glass insulator image, so as to realize the positioning and recognition of glass self exploding insulator. This algorithm can meet the requirements of real-time detection in terms of detection speed and accuracy.

Composite insulator [92] : Composite insulator also known as synthesis insulator, its insulating parts are composed of glass fiber resin core rod, organic material sheath and umbrella skirt. It's characterized by small size, light weight, high tensile strength and excellent pollution flashover resistance, but its anti-aging ability is not as good as porcelain and glass insulators [93]. With the long-term use of composite insulators, there may be some defects. One is external defects that can be seen directly, such as umbrella skirt cracking, core rod exposure, etc.

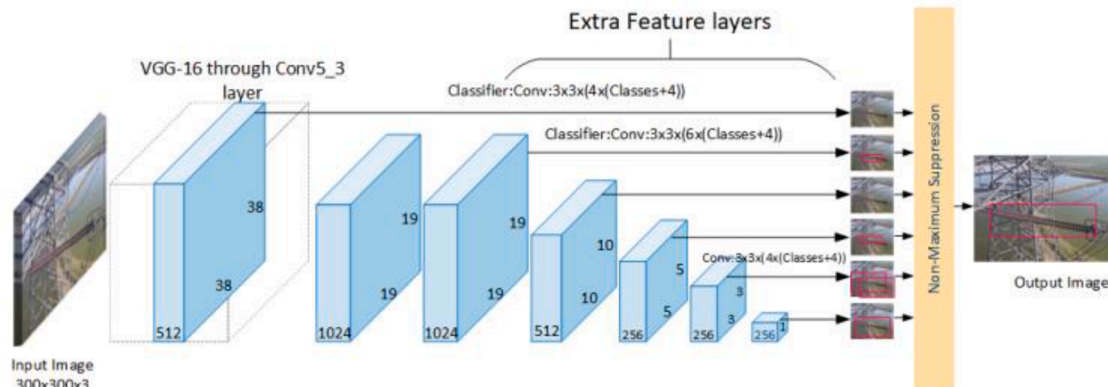


Fig. 9. SSD algorithm structure diagram.



Fig. 10. large insulator detection effect based on deep learning.

The other is internal defect of insulator, such as bubble inside and crack on corerod. Due to the internal partial discharge, external corona discharge and surface resistivity reduction of composite insulator, the local temperature of composite insulator will rise, and the internal defects will lead to serious consequences. Therefore, it is very important to detect the composite insulator regularly. Palangar et al. [94] determined whether a composite insulator was faulty by detecting the leakage current (LC), electric potential (EP) and electric field stress (EFS) of the composite insulator. However, this method requires high equipment requirements and is more complicated to operate. Therefore, Mei et al. [95] proposed a method to detect the internal defects of composite insulators based on microwave technology, and established a full-size automatic detection system for composite insulators on this basis. Through test comparison, the system has high efficiency and stability.

Since the types of insulators in different regions and the environments they are located in may vary in appearance, length, and defect types, and also the defect sites are completely random and will not have exactly the same defects, the detection effects of the model will be different and have some limitations. To enable the model to detect more types of insulator defects, Zhai et al. [96] established a detection model through the unique colors of glass and ceramic insulators, which was based on the spatial morphological characteristics of insulators to complete the detection of dropped string defects in glass and ceramic insulators. Although this method can effectively detect two types of defective insulators, its network structures are all more complex and require more time.

The characteristics and detection methods of different insulators are shown in Table 4.

4.2. Insulator datasets

The insulated datasets contains fewer samples, especially those with defects. The number, quality and diversity of the training samples will affect the performance of the model, and a good model usually cannot be achieved without high-quality training samples. However, as an important national infrastructure, most of the aerial images of insulators are not publicly available for reasons of confidentiality, and only a limited number of public data sets are used for related research, among which the number of defective insulator samples is even smaller, so it is very difficult to collect a large number of high quality insulator samples to train a good model under normal circumstances, and to It is very difficult to collect a large number of high quality insulator samples to train a model with good performance, and it is also very laborious to annotate these data one by one manually. The following are several public data sets collected and compiled for insulator inspection in this section, as shown in Table 5.

CPLID [97] (Chinese Power Line Insulator Datasets), the datasets is an aerial image of high-voltage insulators provided by the State Grid Corporation of China. The datasets contains 600 images of normal insulators as well as 248 synthetic images of self-detonation defective insulators, which can be used for the study of detecting defective insulators.

IDID [98] (Insulator Defect Image Detection), a datasets that focuses on insulator strings as the main object of study, including flashover insulators, insulators with broken shells, and normal insulators. There are 1600 training datasets and 88 test datasets, which can be used for the study of detecting insulators and their defects.

ID (Insulator Datasets) [99], which mainly includes images of pillar insulators under different outdoor lighting and different background conditions, with a total of 2630 images, can be used for the study of insulator detection.

OPDL (Overhead Power Distribution Lines) [100], which is a datasets of insulator images at 15 kV, contains four types of insulators and their defects, including ceramic pin insulators (CPI), ceramic bicolor insulators (CBI), polymer gray insulators (PGI) and green glass insulators (GGI), with a total of 2560 images, which can be used for the study of detecting insulators and their defects.

To solve the problem of small number of samples, we can expand the data sets and adjust the classifier network.

There are two main ways to expand data sets:

- 1) Conduct data enhancement operations on a few types of samples to expand the number of samples. She et al. [101] used data enhancement to expand the original small sample of only 200 insulator images to 3000, and built a VGG16 transmission network, which improved the performance and efficiency of the model through freezing training and fine-tuning training. Reference [102] expands the insulator defect dataset using affine transformation and mosaic, and then completes the defect detection of insulators by the improved YOLOv4 algorithm. However, these simple data enhancement methods do not improve the recognition performance of insulator defect detection algorithms very well.
- 2) Generate sample data by generating adversarial network (GAN) [103]. GAN was proposed by Ian J. Goodfellow and Bengio in 2014. It is mainly composed of generation model and discrimination model. These two models compete with each other and can analyze, capture and copy changes in data sets. Compared with data enhancement through a simple physical approach, GAN not only expands the number of datasets, but also increases the diversity of datasets, making the model more robust. Zhang et al. [104] first segmented insulator images into small pieces using CDSNets segmentation network, and then generated defect samples from these

Table 3

Comparison between traditional detection methods and deep learning detection methods.

Reference	Method	Advantage	Disadvantage
[38]	Insulator segmentation is achieved by texture and shape features.	Insulator can be separated under clutter background.	The shape driven curve generation process and detection is slow.
[48]	Optimized texture features and established a convex energy functional.	It greatly improves the efficiency and accuracy of the algorithm.	The experimental data are single and unconvincing.
[52]	A new feature pyramid is generated by multi-scale feature map, and the target height width ratio of the datasets is calculated by clustering algorithm.	It can overcome the complex background interference of substation and accurately detect the status of insulators.	The method is influenced by the ambient temperature and is prone to false detection.
[64]	The convolutional neural network (CNN) is used to extract visual features from aerial images.	It has strong robustness and applicability.	The detection speed is slow and cannot be realized in real time.
[65]	Improved the RPN and NMS of the model with self-built datasets.	Improved detection accuracy and better detection for shaded insulators.	The accuracy of the detection is not high and the dataset sample is single.
[69]	The vertical projection technology is used to realize the positioning and segmentation of insulators.	High recall and accuracy for broken, fouled, foreign matter and flashover insulators.	The network structure is more complex and the detection speed is slow.
[71]	The improved YOLOv3 algorithm is used, and EIoU is used as the loss function.	It can quickly and accurately detect insulators and their defects.	The mAP value of the defective insulator detection is only 89%.
[74]	Improved YOLOv4 algorithm using light-weighting methods and data augmentation.	Improved inspection speed and accuracy using lightweight methods.	Poor detection for obscured targets and small targets.
[77]	The feature fusion module of YOLOv5 algorithm is replaced by BiFPN network, and CA attention mechanism is added.	The average precision of insulator defect detection is 89.1%, which is 2.7% higher than the original algorithm.	The network model is complex and the detection results are not accurate enough.
[81]	Improving the loss function of the model and introducing the ECA attention mechanism.	The improved model can accurately and real-time detect insulators and their defects.	Increases the number of parameters and computational of the model.
[83]	An improved SSD algorithm is proposed, which can automatically extract the deep features of insulators from images.	It allows fast and accurate identification of porcelain and composite insulators in complex environments.	The model structure is complex and the detection speed is slow.

small pieces by the proposed DefGANS, thus enriching the number of insulator defect samples and finally completing defect detection. However, the technology of insulator defect sample generation based on GAN is not yet mature, which cannot solve the problem of insufficient insulator defect samples. Based on this, Reference [105] proposes an insulator sample expansion method based on the cyclic consistent generation adversarial network (Cycle GAN), and the samples generated by this method can replace the real samples to a certain extent. On the basis of this network, Reference [106] increased the diversity of samples by changing the loss function model. Compared with other amplification methods, this method has significantly improved the accuracy of insulator defect detection.

For the problem of annotation of datasets, there are two main approaches.

- Firstly, A small number of labeled vertebral samples were used. The algorithm was improved so that the algorithm can have good detection even with a small number of labeled samples. Zhang et al. [107] investigated the defect detection method for insulators with a small number of labeled samples using terahertz wave devices and obtained high accuracy and recall.
- Secondly, Automatic generation of labels for samples using semi-supervised learning. Wang et al. [108] proposed a perspective transformed data augmentation approach (PTDA), which can automatically label newly generated data without expert knowledge, not only increasing the weight of rare samples in the datasets, but also improving the robustness of the model.

4.3. Defect type and cause

There are many defects in insulators, and the power grid is vulnerable to various types of defects. It is difficult for the training set to cover all types of defects. The lack of training sample data and the limitations of deep learning model in predicting unknown types pose great challenges to insulator defects. The detect types and causes of insulators are mainly as follows:

Self explosion of insulator [109]: The main causes of insulator string fracture are external tearing, brittle fracture (stress corrosion) and perishable fracture, where external tearing cases rarely occur, mainly

brittle fracture [110] and perishable fracture. External tearing is generally caused by high winds, severe ice cover on the line, etc., in which case the insulator is subjected to forces beyond its limits, leading to fracture. The mandrel of early insulators is a non-acid-resistant material, so brittle fracture is the main cause of early insulator string fracture. The brittle fracture is generally accompanied by a stress corrosion process. There are two main conditions that lead to brittle fracture, i.e. the insulator seal fails and the core rod is non-acid resistant. In addition, the generation of cracks on the insulator surface can also greatly accelerate the occurrence of brittle fracture. Decay is often the result of long term discharge inside the mandrel causing it to fracture due to its own inability to withstand the load. If not replaced in a timely manner, transmission lines due to reduced creepage distance is very vulnerable to single-phase grounding or short-circuit faults, and even serious safety incidents leading to large scale power outages in the power grid system [111,112]. At the same time, the self-detonation of insulators causes a part of the information to be lost, which makes the detection process a bit difficult. In view of this, Zhang et al. [113] designed an adaptive receiver network (ARFNet) that can be used in almost all two-stage detection networks, enabling the network to detect the background information of insulator self-detonation and thus improve the accuracy of insulator self-detonation defect detection. For example, Ling et al. [114] first combined the Faster R-CNN algorithm with the U-net algorithm to complete the defect detection of self bursting glass insulators at low signal-to-noise ratio.

Insulator pollution flashover [115]: Insulator flashover refers to the phenomenon of discharge along the surface of a solid insulator when the gas or liquid dielectric around the solid insulator is penetrated. The voltage at its discharge is called the flashover voltage. After phashover, the voltage between the electrodes rapidly drops to zero or near zero. The spark or arc in the flashover channel can heat the insulation surface, cause carbonization and damage the surface insulation. As insulators are exposed to the external environment for a long time, they are affected by carbon dioxide, dust and bird droppings, which can form a fouling accumulation on their surfaces. Normally, if the insulator surface is kept smooth and dry, then this dirt does not cause insulator flashover or discharge along the surface. However, when the dirt is affected by the natural environment such as light rain and dew and becomes wet, a conductive layer is created on the surface, which greatly increases the leakage current. This process is known as contamination flashover [116,

Table 4

Different types of insulators and their detection methods.

	Advantage	Disadvantage	Common defects	Detect methods
Ceramic insulator	1 Good insulation performance. 2 Large mechanical strength. 3 Long service life. 4 Strong adaptability. 5 Convenient operation and maintenance.	1 Poor temperature resistance. 2 Defects are difficult to be found. 3 High maintenance costs.	1 Breakdown. 2 Flashover. 3 Surface rupture and corona discharge.	An approach combining infrared images with convolutional neural networks [86]. Use ultraviolet image sample library [87], Use low light level image enhancement [88]
Glass insulator	1 Large mechanical strength. 2 Strong anti-aging ability. 3 low operation and maintenance cost. 4 zero value self explosion, easy to detect.	Due to its small size and weight, it is easy to explode.	Self explosion and string breaking.	According to image characteristics [89–91].
Composite insulator	1 Small size and light weight. 2 easy maintenance. 3 Good anti pollution performance. 4 strong corrosion resistance.	1 High production cost. 2 Weak anti-aging ability. 3 Insulator cap is soft and easy to break.	1 Lightning flashover. 2 pollution flashover. 3 external damage.	Using microwave technique and leakage current method [94,95].

Table 5

Transmission line insulators open data set.

Datasets	Size	Research Scenarios	Download Links
CPLID [97]	848	Insulators and their self-destruction defects detection.	https://github.com/InsulatorData/InsulatorDataSet
IDID [98]	1688	Insulator breakage, fouling flash, corrosion defect detection.	https://dx.doi.org/10.21227/vkdw-x769
ID [99]	2630	Insulator detection.	https://cv.po.opole.pl/dataset1
OPDL [100]	2560	Insulator type classification and defect detection.	http://www.dee.eng.ufba.br/dslab/index.php/opdl_dataset

[117] due to the increase in local current, which leads to discharge on the insulator surface.

The leakage current in the flashover process is divided into three stages such as safety stage, prediction stage and danger stage, etc. The timely detection of anomalies in the safety stage is the most effective period to prevent the insulator from contamination flashover [118]. Leakage current is one of the most important and effective causes of insulator pollution flashover because it reflects the condition of the insulator and the distance between the insulator and the flashover [119]. Jose-M [120] et al. designed a maintenance plan for high-voltage insulators to prevent contamination flashover of insulators in order to improve the reliability and safety of the grid. For example, Zhai et al. [121] proposed a detection method of porcelain insulator flashover failure based on significance and morphological techniques, and determined the defect location by using the color characteristics of the insulator damaged area.

Rust or falling off of insulator steel cap: Due to the material problem of insulator and long-term exposure to complex external environment, the accumulation of insulators over time leads to rusting and even detachment of insulator steel caps. For example, Faiyaz Ahmed et al. [122] focused on the development of autonomous UAVs with deep learning methods as the backbone analysis and a new insulator datasets, which is capable of detecting defects such as cracks, missing and broken steel caps of insulators. Although this method improves the accuracy of detection, it requires annotation of a large number of insulator samples, so Shi et al. [123] proposed a weak supervision method, only to the insulator boundary box annotation, the method saves some work, and can detect insulators and missing cap from the arbitrary angle of the image.

Insulator icing [124]: In extreme weather, such as snow storm and low temperature, it will lead to insulator icing, and when the ice melts, it will also form a conductive layer on the insulator surface, leading to

insulator flashover. Flashover [125] is also a relatively common fault that endangers the safe operation of the power grid and causes large-scale power outages, resulting in huge economic losses. Therefore, it is particularly important to detect insulator icing. Yang et al. [126] established a method that can identify insulator icing types based on texture features of insulators, and summarized six common types of insulator icing by extracting texture features from typical images of six types of ice and snow, including ice and snow, heavy frost, medium frost, light frost, partial frost, and snow. Zhu et al. [127] studied the morphological characteristics of ice-covered insulators based on the effect of insulator ice-covered flashover, and also compared the detection methods of ice-covered insulators in recent years. The study showed that the image detection technology has good effect for the detection of ice-covered insulators, and finally proposed a good research background based on deductive method and machine learning method for the detection of ice-covered insulators. Fig. 11 Insulator self-explosion, flashover, corrosion and falling of steel cap and icing defects.

Reference [128] enhanced the detection of minor defects using improved ResNeSt and RPN and was able to accurately detect four types of insulators as well as three different defect types. Due to the different detection effects of the algorithm on different types of insulators, there are certain limitations, which reduces the detection performance of the algorithm. Therefore, Wang et al. [129] proposed a cascade detection method, in the first stage, the model can detect more complex and subtle defects through image clipping. In the second stage, a multi-scale feature re-weighting network is established to detect defective insulators, and the proposed two-stage structure can detect five common insulator defects. Huang et al. [130] uses multiple model training of various types of aerial images of insulators, and then classifies and identifies the defect types. This method can identify multiple defects in the insulator image, which improves the accuracy of insulator fault detection. Although the above methods can effectively detect various types of defective insulators, their network structures are more complex and require more time.

5. Challenges and future development direction

5.1. Challenges

At present, the method based on deep learning has made great achievements in the field of insulator defect detection, and has become the mainstream algorithm for detecting defective insulators. However, there are still some difficult problems hindering the development and progress of this field.



Fig. 11. Insulator self-explosion, flashover, corrosion and falling of steel cap and icing defects.

5.1.1. Complex network model

Although deep learning-based detection methods have greatly improved the speed of detection compared with manual detection as well as traditional detection methods, the current defective insulator detection algorithm has a complex structure, number of parameters, large computation and other problems, and the way of uploading pictures to the cloud for detection through drones still cannot meet the demand for real-time detection.

5.1.2. Small target detection is not effective

Small target defect detection is not effective. Due to electromagnetic interference, UAVs need to maintain a safe distance from transmission lines during inspection, while the defective parts of insulators are much smaller compared to normal equipment (e.g., the area where defects occur in insulators compared to intact insulator strings), which will lead to a situation where the algorithm may overlook feature extraction from these defective areas, resulting in missed detections, which poses a challenge for defect detection.

5.1.3. Feature extraction in complex backgrounds

Insulators are usually in responsible environments such as high mountains, rivers, farmlands, construction sites, forests, etc. The UAV cannot achieve all-round coverage during the filming process, and the images of insulators may be obscured by other power equipment, and there are problems such as unclear images, which make it difficult for the algorithm to effectively extract the characteristic information of insulators and their defects, thus affecting the accuracy of detection and leading to leakage and false detection.

5.2. The future direction of development

In the field of insulator defect detection, the deep learning-based method has the advantages of lower cost, simpler operation, faster detection speed and higher efficiency compared with other methods, and it has achieved significant results in the insulator defect detection

task, but at the same time, the algorithm also has some problems, such as complex network model, image feature extraction, small target detection and too dependent on data set, etc. In view of the above problems, future research can be conducted in the following aspects.

5.2.1. Complex network model

To simplify the network model, scholars have proposed lightweight network structures [131] as well as edge computing (EC) [132,133].

Current light-weighting methods for deep learning networks are mainly achieved by changing the network structure, reducing the complexity of the model and reducing the number of parameters, for example, deep separable convolution (DSConv) can be used instead of traditional convolution to reduce the convolution kernel parameters and the number of floating point computations, and DSConv consists of deep convolution (DConv) and point-by-point convolution (PConv). The main commonly used lightweight networks are Mobile-Net, Shuffle-Net, Ghost-Net and Efficient-Net, for example, Xia [134] et al. used Mobile-Net V1 to replace the feature extraction network of Center-Net model and introduced both CBAM module and transposed convolution module, which improves the accuracy and at the same time reducing the complexity of the model and reducing the model size by 37 MB. Wang et al. [135] used Efficient-Net lightweight network to replace the feature extraction network of the original algorithm while adding the attention module to maintain the detection accuracy, and the model size was reduced by 45.8%, making it easier to deploy in embedded devices such as UAVs. Lan et al. [136] used the Efficient-Net lightweight network to replace the feature extraction network of the original algorithm while adding the attention module to maintain the detection accuracy, and the model size was reduced by 45.8%, making it easier to deploy in embedded devices such as UAVs. YOLOv5 model by introducing Ghost module and also improving the model, which effectively reduces the complexity of the model and improves the speed of detection while maintaining the accuracy.

Edge computing does not require long distance data transmission and enables local processing of distributed data, avoiding centralized

processing of large-scale data, but the method is limited by computational and storage capacity, the processing speed is slow and the detection accuracy is not high, which is not suitable for deployment in large network models. So Song et al. [137] combined cloud computing and edge computing for defect detection of insulators in industrial Internet of Things (IIoT), while reducing the UAV computation, and they were the first to analyze the impact of the great aspect ratio of insulators on the detection accuracy as well as computation. Deng et al. [138] proposed a fused mobile edge computing that reduces the network structure of YOLOv4, while using deep neural network division algorithm makes full use of the computational resources of UAVs.

The current improvement of insulator defect detection algorithm generally improves detection accuracy at the cost of increasing computational complexity, and there is less improvement for network light-weighting. It is important to conduct research on algorithm light-weighting for edge computing devices to make them easier to deploy in mobile, so as to realize real-time detection during inspection.

5.2.2. Small target detection

For the problem that defect targets are relatively small and not easy to detect, a multi-stage detection method can be used to divide the defect detection of insulators into two steps [139]: the first step first identifies and segments the relatively large scale targets, and the second step uses a defect point detection model with finer detection granularity to detect the segmented area. Zhao et al. [140], in order to improve the accuracy of insulator identification and defect detection, first use FPN improved Faster R-CNN model to locate insulators in complex background images, then perform image segmentation on target areas, and finally complete insulator defect detection. In order to extract defect features of micro insulators in complex environments, Li et al. [141] proposed a cascade detection and segmentation network method to identify defects from both global and local levels. This method greatly reduces the rate of missed detection and has strong robustness and accuracy.

The multi-stage detection method improves the percentage of defective regions of insulators in the image, solves the problem of difficult detection of small targets from the input side of the model, and has a certain improvement compared with single-stage detection. However, this type of method also has obvious disadvantages. Firstly, it requires cascading multiple deep learning networks, so its network model parametric number and computational complexity are relatively high, which leads to a slower detection speed and higher requirements for computing equipment. Secondly, multi-stage detection will lead to error transfer from the training error of the previous network model to the next one due to the increase of detection steps.

5.2.3. Feature extraction

In order to better extract the defect features of insulators, the usual method is to add attention mechanism [142], whose main role is to enhance the extraction of various appearance features for the target, making the algorithm biased to extract that feature. Its core is to let the network focus on the area of the target in the image, rather than the entire image. In recent years, attention mechanism has been widely used in computer vision tasks such as target detection and image segmentation, and has occupied an important position in the field of deep learning, by making the algorithm focus on the feature information of the target, while ignoring other unimportant information, and improving the detection performance of the algorithm. Huang et al. [143] enhanced the semantic information of the feature map by adding a lightweight attention mechanism ECA to the lightweight YOLOX-tiny model, which provides better detection for small target insulators as well as obscured insulators. For example, Hu et al. [144] introduced the CBAM attention mechanism in the skip connection structure to make the network more focused on the extraction of feature information, and reduced the segmentation error rate as well as the detection problem of insulator ice cover region using the proposed SGAN-UNet. Chen et al. [145] extracted channel feature information by AF module and solved

the problem of spatial feature information loss, while using DSP module to improve the detection performance of the model in complex background.

In addition, it can be improved in the following three ways:

- Firstly, Improve the quality of acquired images. To provide image quality, Chen et al. [146] used SRCNN to perform super-resolution reconstruction of blurred images and then expanded the dataset by combining it with other training images. Li et al. [147] and others proposed an image enhancement method based on illumination correction and compensation in order to solve the problem of unclear insulator images, and corrected it with a two-dimensional adaptive gamma transform, which makes the images have a strong noise immunity.
- Secondly, Defects of insulators are detected by fusing infrared, ultraviolet, laser radar and other sensors. For example, Gao et al. [148] proposed a batch normalized convolution attention module (BN-CBAM) and feature fusion module to solve the problem of low detection accuracy caused by the defective part of insulator is small in the patrol image. This method has better effect than other methods in small insulator defect detection. Mussina et al. [149] proposed a new fusion convolutional network (FCN) architecture and developed a multimodal information fusion (MMIF) system, which reduced the complexity of the CNN structure and enabled the FCN system to provide higher classification accuracy with less time and memory consumption.

6. Conclusion

At present, with the continuous progress of depth science method and the update iteration of computer equipment, it has achieved remarkable results in the field of target detection and has been skillfully applied in the fields of defect detection, fault diagnosis and unmanned driving. As one of the most important infrastructures in China, transmission lines play an important role in the transmission and distribution process of the power system, supplying the electricity needs of various regions in China. And as insulators are the most numerous and highest failure rate components in transmission lines, fast and accurate identification of defective insulators is of great significance for the safe and stable operation of transmission lines.

Compared with other detection methods, the deep learning-based method is less demanding on equipment and simpler and easier to deploy, and has an irreplaceable position for insulator defect detection. This paper firstly introduces the early insulator defect detection methods, and secondly details the defective insulator detection method using UAV as the inspection tool, aerial insulator images as the data set, and deep learning algorithms as the main detection means, and systematically introduces the development history of insulator defect detection as well as the challenges and future development directions. At the same time, the method described in this paper can be used not only for the detection of defective insulators, but also for the fault detection of other components in transmission lines. Traditional detection methods need to extract image features or temperature changes of insulators. Such methods need to segment the image first, extract the surface feature information of the image, and then judge whether there are defects according to the features designed manually. However, most insulators are located in farmland, mountain, river, forest and other complex environments, and traditional detection methods cannot detect defective insulators well. With the continuous progress and development of deep learning, target detection technology has become more and more perfect and has been applied in many fields. In the image field, the target detection algorithm can achieve end-to-end operation, greatly shorten the detection time, and can accurately identify the area where the target is located. At present, target detection algorithms based on deep learning are mainly divided into two stages and one stage, and they have achieved good results in the field of insulator defect detection.

However, there are still some key technologies that need to be broken through:

- Firstly, Small sample learning method. Image enhancement is an effective means to expand the number of training samples. Commonly used data enhancement methods include clipping, mirroring, adding noise, etc. Physical rendering method and ray tracing technology can effectively increase the number of samples. In the case of limited number of samples, how to use small sample data to train a good deep learning network model will be one of the research contents of insulator defect detection in the future.
- Secondly, Establish a unified data set and evaluation criteria. Due to the particularity of the transmission line industry, the primary problem facing the application of deep learning in the target detection of power equipment is the lack of open data set, which requires the support of a large number of sample data. The construction of open data set of insulator defects can greatly promote the implementation of deep learning technology. Secondly, due to the differences among targets and the diversity of algorithms, there is still a lack of unified evaluation standards for power equipment identification and defect detection, so it is urgent to establish an evaluation standard for insulator defect performance.
- Thirdly, Multi-level detection of small targets. Small target defect detection of insulators, such as cracks, accounts for more than 50%. Object detection, image classification and semantic segmentation algorithm are combined to form an object detector using prior knowledge, joint features, context information, etc., through the step by step detection and segmentation of restricted targets, multilevel hierarchical detection strategy will be an effective solution for small target defect detection.
- Fourth, Multi-layer feature fusion. Under the framework of deep learning network, shallow convolution is used to extract features for more texture and shape features, while deep convolution is used to extract features for higher semantic features. With the deepening of the layers of deep learning network, texture and shape information of detection targets are weakened and corresponding semantic information is continuously enhanced. Feature pyramid structure is introduced to carry out multi-scale feature deep fusion. Strengthen the feature extraction ability of detection target.
- Fifth, Multi-source image data fusion. Using traditional image processing methods to enhance the image (such as image graying, filtering, histogram equalization and image segmentation, etc.) can effectively avoid the degradation problem of acquired aerial image. In addition, the multi-source data fusion of visible image, thermal image and ultraviolet image can effectively expand the application range of insulator defect detection and realize insulator defect detection.
- Sixth, The insulator defect detection model is complex and difficult to deploy in drones, mobile robots and other equipment to achieve real-time detection. Lightweighting the network to reduce the number of parameters as well as the amount of computation, or using edge computing methods is important for real-time detection.

Finally, Deep integration of multiple learning methods. In view of the limited number of samples in the data set, a small part of sample data is adopted to train on the well trained model, and the advantages of different learning technologies are complementary through the deep integration of deep learning with transfer learning, weak supervised learning and reinforcement learning. The continuous integration of deep learning and various machine learning technologies will be the research focus of the application of deep learning in transmission line inspection in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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